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(54) **NON-ASPHALTIC COATINGS,
NON-ASPHALTIC ROOFING MATERIALS,
AND METHODS OF MAKING THE SAME**

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None

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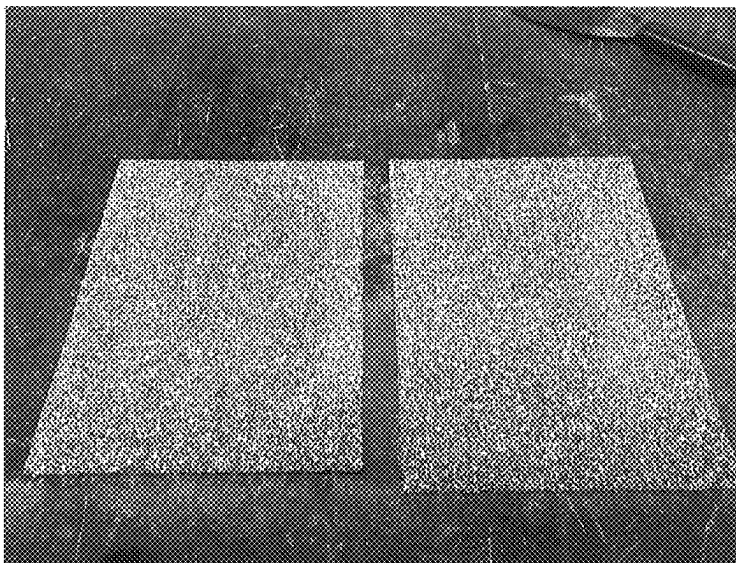
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(57) **ABSTRACT**

Non-asphaltic coatings for roofing materials, roofing materials made therefrom and methods of preparing such coatings and roofing materials utilize a combination of a crude tall oil-based continuous phase material, a resinous hardening agent, and a polymer.

11 Claims, 2 Drawing Sheets



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Fig. 1

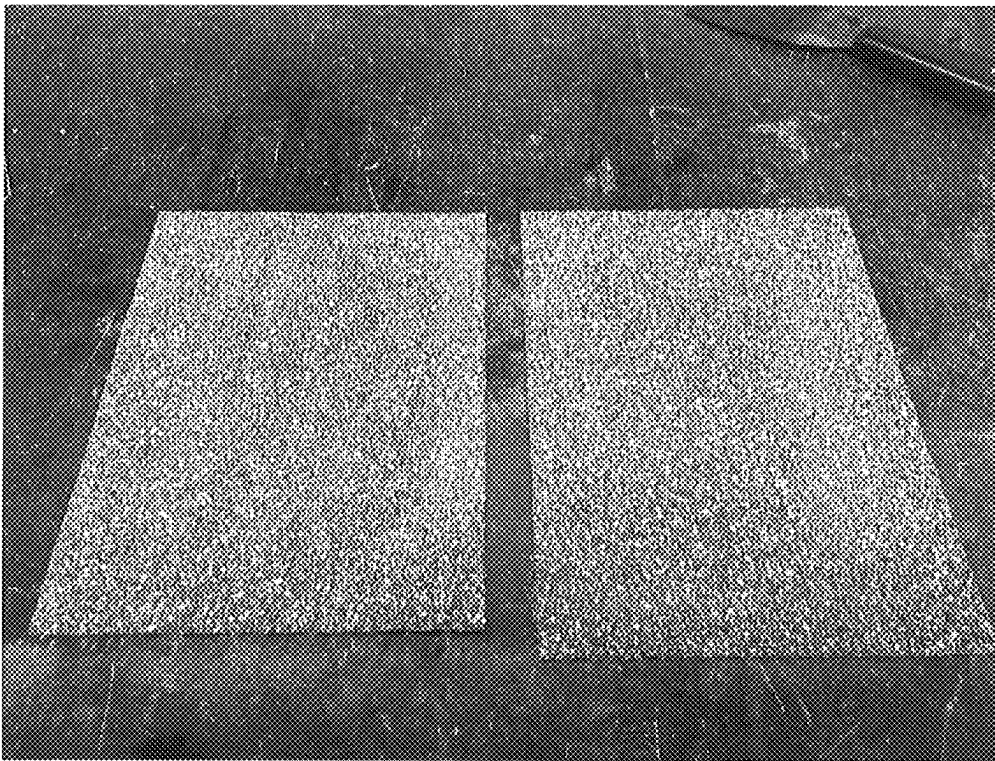
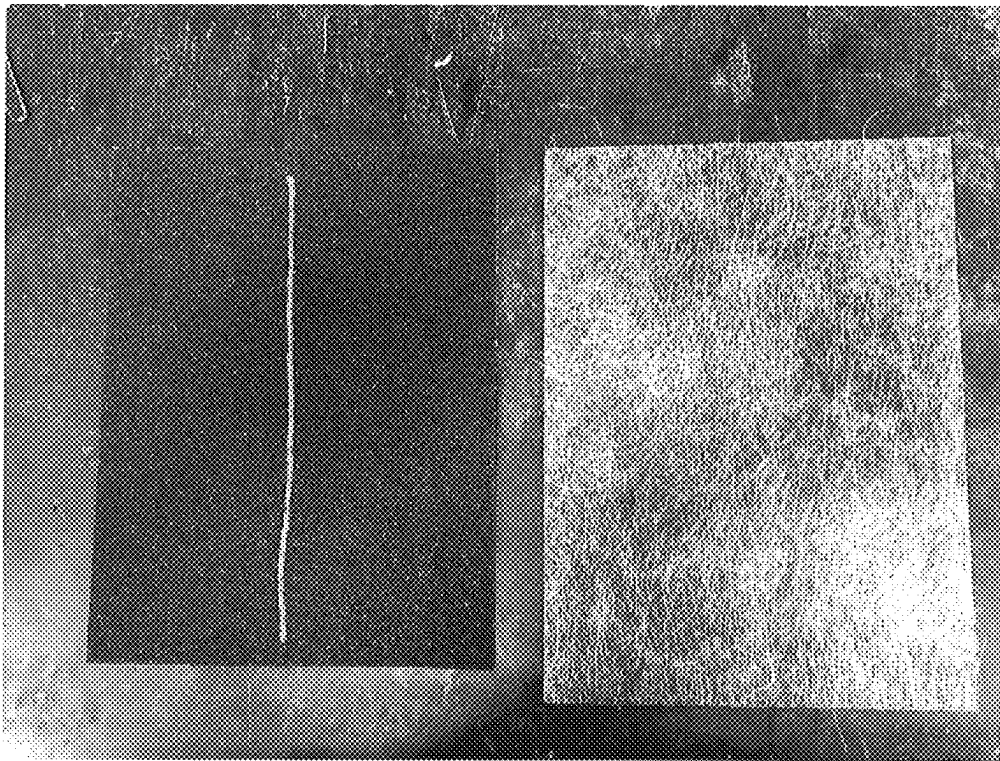


Fig. 2



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NON-ASPHALTIC COATINGS, NON-ASPHALTIC ROOFING MATERIALS, AND METHODS OF MAKING THE SAME

RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/400,827 filed Aug. 12, 2021, now granted as U.S. Pat. No. 11,466,176, which claims the priority of U.S. provisional application No. 63/065,763, filed August 14, 2020, and U.S. provisional application No. 63/121,447, filed Dec. 4, 2020, which are incorporated herein by reference in their entirety for all purposes.

BACKGROUND OF THE INVENTION

This disclosure relates to non-asphaltic coatings for roofing materials, to roofing materials made therefrom and to methods of preparing such coatings and roofing materials. A combination of a crude tall oil-based continuous phase material, a resinous hardening agent, and a polymer is used in the non-asphaltic coatings. Roofing materials, such as, e.g., shingles, using these coatings or made by these methods have comparable or superior properties to roofing materials having a traditional asphaltic coating.

Traditional roofing materials, such as, e.g., shingles, are based upon a glass or felt mat that is coated and impregnated with an asphalt-based composition, e.g., an oxidized asphalt or polymer-modified asphalt (PMA), that is subsequently coated with granules.

SUMMARY OF THE INVENTION

An embodiment of the disclosure relates to a filled coating formulation comprising:

- 5-40 wt % of a crude tall oil-based continuous phase material;
- 3-35 wt % of a resinous hardening agent;
- 1-10 wt % of a polymer; and
- a filler;

wherein a viscosity of the filled coating formulation ranges from 2000-20000 cP measured at 400° F.

In an embodiment, the crude tall oil-based continuous phase material comprises tall oil pitch, tall oil pitch blends, distilled tall oil, tall oil pitch blends with low sterols, crude tall oil, or a combination thereof

In an embodiment, the resinous hardening agent comprises esters of maleated rosin with pentaerythritol or stabilized pentaerythritol ester of rosin-based tackifier; rosin, maleated, polymer with glycerol; resin acids and rosin acids, esters with pentaerythritol; modified rosin ester; pentaerythritol ester of rosin; alpha methyl styrene tall oil resin; or a combination thereof

In an embodiment, the polymer comprises poly(styrene-butadiene-styrene) (SBS), poly(styrene-ethylenebutylene-styrene) (SEBS), ethylene-vinyl acetate copolymer (EVA), polyvinyl butyral (PVB), or a combination thereof

In an embodiment, the filler comprises calcium carbonate, barium sulfate, calcium sulfate, talc, limestone, perlite, silica, fumed silica, precipitated silica, quartz, aluminum trihydrate, magnesium hydroxide, colemanite, titanium dioxide, fly ash, graphene nanoparticles, carbon black, recycled rubber tires, recycled shingles, recycled thermoplastic resins, basalt, roofing granules, clay, lignin, or a combination thereof.

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In an embodiment, the filled coating formulation further comprises a dye, a pigment, a fire retardant, a UV stabilizer, or a combination thereof.

In an embodiment, the filled coating formulation comprises 40-75 wt % of filler.

An embodiment of the disclosure relates to a non-asphaltic roofing shingle comprising a substrate having a coating which comprises a crude tall oil-based continuous phase material, a resinous hardening agent, a polymer, and a filler, wherein the softening point of the coating ranges from 175-320° F., and

wherein the penetration of the coating ranges from 5-100 dmm measured at 77° F.

In an embodiment, the substrate comprises a fiberglass mat, a polyester mat, a scrim, a coated scrim, or a combination thereof.

In an embodiment, the non-asphaltic roofing shingle is configured to be prepared on a substantially standard manufacturing line for asphaltic shingles at a standard speed.

In an embodiment, the non-asphaltic roofing shingle is a single layer shingle or a laminated shingle having two or more layers.

In an embodiment, a thickness of the coating on the substrate ranges from 5-100 mils.

In an embodiment, the non-asphaltic roofing shingle comprises one or more layers of the coating.

In an embodiment, the non-asphaltic roofing shingle further comprises granules.

An embodiment of the disclosure relates to a method comprising:

- obtaining a substrate; and
- coating the substrate with a filled coating formulation to form a non-asphaltic roofing material,
- wherein the filled coating formulation comprises 5-40 wt % of a crude tall oil-based continuous phase material, 3-35 wt % of a resinous hardening agent, 1-10 wt % of a polymer, and a filler,
- wherein a viscosity of the filled coating formulation ranges from 2000-20000 cP measured at 400° F.

In an embodiment, coating the substrate is performed on a substantially standard manufacturing line for asphaltic shingles at a standard speed.

In an embodiment, the method further comprises applying granules to the non-asphaltic roofing material.

In an embodiment, the method further comprises applying one or more layers of the coating to form the non-asphaltic roofing material.

In an embodiment, the method further comprises preparing the coating.

In an embodiment, the method further comprises cutting the formed non-asphaltic roofing material to provide one or more roofing shingle layers and configuring the one or more roofing shingle layers to provide a shingle having a desired structure.

In an embodiment of the method, the filled coating formulation comprises 40-75 wt % of filler.

An embodiment of the disclosure relates to a non-filled coating formulation comprising:

- 20-60 wt % of a plant- or bio-derived continuous phase material;
- 30-75 wt % of a resinous hardening agent; and
- 5-20 wt % of a polymer,
- wherein a viscosity of the non-filled coating formulation ranges from 200-12,500 cP measured at 400° F.

In an embodiment, the plant- or bio-derived continuous phase material comprises corn oil, vegetable oil, triglycerides, or a combination thereof.

BRIEF DESCRIPTION OF THE FIGURES

For a more complete understanding of the invention and the advantages thereof, reference is made to the following descriptions, taken in conjunction with the accompanying figures, in which:

FIGS. 1 and 2 are photographs of the front and back surfaces, respectively, of coupons of a filled coating formulation according to an embodiment of the disclosure.

DETAILED DESCRIPTION

The present disclosure provides non-asphaltic coatings for roofing materials, roofing materials made therefrom and methods of preparing such coatings and roofing materials. The disclosure utilizes a combination of a crude tall oil-based continuous phase material, a resinous hardening agent, and a polymer. This combination can be used with existing equipment for the manufacture of shingles and similar roofing material with minimal process changes. In addition, to a significant extent, the materials employed herein are biologically derived and are by-products of the paper-making process.

Among those benefits and improvements that have been disclosed, other objects and advantages of this disclosure will become apparent from the following description taken in conjunction with the accompanying figures. Detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely illustrative of the disclosure that may be embodied in various forms. In addition, each of the examples given regarding the various embodiments of the disclosure are intended to be illustrative, and not restrictive.

Throughout the specification, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise. The phrases “in one embodiment”, “in an embodiment”, and “in some embodiments” as used herein do not necessarily refer to the same embodiment (s), though they may. Furthermore, the phrases “in another embodiment” and “in some other embodiments” as used herein do not necessarily refer to a different embodiment, although they may. All embodiments of the disclosure are intended to be combinable without departing from the scope or spirit of the disclosure.

As used herein, the term “based on” is not exclusive and allows for being based on additional factors not described, unless the context clearly dictates otherwise. In addition, throughout the specification, the meaning of “a”, “an”, and “the” include plural references. The meaning of “in” includes “in” and “on”.

As used herein, terms such as “comprising”, “including,” and “having” do not limit the scope of a specific embodiment to the materials or steps recited by the embodiment.

All prior patents, publications, and test methods referenced herein are incorporated by reference in their entireties.

As used herein, the term “free of asphalt” means that the coating does not include any amount of asphalt. In other words, the coating includes 0% by weight of asphalt.

As used herein, the terms “composite” and “coated substrate” are used interchangeably.

As used herein, the term “weight percent” means the percentage by weight of a component based upon a total weight of the filled coating formulation, composite or coated substrate, as applicable.

One embodiment of the disclosure provides a filled coating formulation comprising:

5-40 wt % of a crude tall oil-based continuous phase material;

3-35 wt % of a resinous hardening agent;

1-10 wt % of a polymer; and

a filler,

wherein a viscosity of the filled coating formulation ranges from 2000-20000 cP measured at 400° F.

Non-limiting examples of crude tall oil-based continuous phase materials used in the present disclosure include, without limitation, tall oil pitch, tall oil pitch blends, distilled tall oil, tall oil pitch blends with low sterols, crude tall oil, and combinations thereof. One of ordinary skill in the art will readily understand what is meant by “low sterols” with regard to tall oil pitch blends with low sterols; for example, “low sterols” may indicate that some amount of sterols has been removed from a given tall oil pitch blend. In an embodiment, the crude tall oil-based continuous phase material is Sylfat DP-1 (Kraton Chemical LLC), Sylfat DP-8 (Kraton Chemical LLC), Dertoline Poix de Tall Oil (Les Derives Resiniques et Terpeniques (DRT)), Altapyne pitch (Ingevity Corporation), Tufftrek 3100 (Georgia Pacific Chemicals), or any combination thereof. As used herein, “crude tall oil-based continuous phase material(s)” refers to any material used or obtained during the distillation of crude tall oil (CTO) including crude tall oil and tall oil pitch (TOP), which is the residue obtained from the distillation of crude tall oil; for purposes of the present disclosure, “crude tall oil-based continuous phase material(s)” also includes materials derived from any material used or obtained during the distillation of crude tall oil. TOP is a dark, tar-like soft solid at room temperature.

In an embodiment, the filled coating formulation comprises 5-40 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 5-35 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 5-30 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 5-25 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 5-20 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 5-15 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 5-10 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 10-40 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 10-35 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 10-30 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 10-25 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 10-20 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 10-15 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 15-40 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 15-35 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 15-30 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 15-25 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 15-20

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wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 20-40 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 20-35 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 20-30 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 20-25 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 25-40 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 25-35 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 25-30 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 30-40 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 30-35 wt % of crude tall oil-based continuous phase material. In an embodiment, the filled coating formulation comprises 35-40 wt % of crude tall oil-based continuous phase material.

Non-limiting examples of resinous hardening agents used in embodiments of the present disclosure include, without limitation, esters of maleated rosin with pentaerythritol or stabilized pentaerythritol ester of rosin-based tackifier; rosin, maleated, polymer with glycerol; resin acids and rosin acids, esters with pentaerythritol; modified rosin ester; pentaerythritol ester of rosin; alpha methyl styrene tall oil resin; and any combination thereof. In an embodiment, the resinous hardening agent is Dertoline P-110 (Les Derives Resiniques et Terpeniques (DRT)), Sylvacote 7097 (Kraton Chemical LLC), Sylvacote RE-98 (Kraton Chemical LLC), WestRez 5110 (Ingevity Corporation), Sylvacote 4984 (Kraton Chemical LLC), Dertoline P-105 (Les Derives Resiniques et Terpeniques (DRT)), Sylvares 115, or any combination thereof. As used herein, "resinous hardening agent (s)" refers to any material composing the rosin fraction obtained during distillation of crude tall oil, e.g., rosin acids, resin acids, etc.; for purposes of the present disclosure, "resinous hardening agent(s)" also includes materials derived from any material composing the rosin fraction obtained during distillation of crude tall oil, e.g., derived by esterification. According to embodiments of this disclosure, a resinous hardening agent having a softening point ranging from 80° C.-140° C. is useful or a resinous hardening agent having a softening point ranging from 80° C.-135° C. is useful. According to embodiments of this disclosure, a resinous hardening agent is used in combination with a crude tall oil-based continuous phase material to raise the softening point thereof to a level that makes the crude tall oil-based continuous phase material suitable for use in a roofing material. It is further believed that the use of a resinous hardening agent herein changes the resistance to deformation of a resulting product, e.g., prevents granules from moving on a roofing shingle.

In an embodiment, the filled coating formulation comprises 3-35 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 3-30 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 3-25 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 3-20 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 3-15 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 3-10 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 3-5 wt % of resinous hardening agent.

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In an embodiment, the filled coating formulation comprises 5-35 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 5-30 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 5-25 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 5-20 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 5-15 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 5-10 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 10-35 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 10-30 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 10-25 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 10-20 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 10-15 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 15-35 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 15-30 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 15-25 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 15-20 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 20-35 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 20-30 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 20-25 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 25-35 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 25-30 wt % of resinous hardening agent. In an embodiment, the filled coating formulation comprises 30-35 wt % of resinous hardening agent.

In an embodiment, the filled coating formulation comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 2000-20000 cP measured at 400° F. In an embodiment, the filled coating formulation comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 2000-15000 cP measured at 400° F. In an embodiment, the filled coating formulation comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 2000-10000 cP measured at 400° F. In an embodiment, the filled coating formulation comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 2000-5000 cP measured at 400° F. In an embodiment, the filled coating formulation comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 5000-20000 cP measured at 400° F. In an embodiment, the filled coating formulation comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 5000-15000 cP measured at 400° F. In an embodiment, the filled coating formulation comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 5000-10000 cP measured at 400° F. In an embodiment, the filled coating formulation comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 10000-20000 cP measured at 400° F. In an embodiment, the filled coating formulation comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 10000-15000 cP measured at 400° F. In an embodiment, the filled coating formulation

comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 15000-20000 cP measured at 400° F.

Non-limiting examples of polymers used in embodiments of the present disclosure include, without limitation, poly (styrene-butadiene-styrene) (SBS), poly(styrene-ethylene-butylene-styrene) (SEBS), ethylene-vinyl acetate copolymer (EVA), polyvinyl butyral (PVB), polystyrene, and any combination thereof. In embodiments, the polymer is branched SBS block copolymer, 30% styrene; radial SBS block copolymer, 32% styrene; linear SEBS block copolymer, 29% styrene; ethylene-vinyl acetate copolymer, 28% vinyl acetate; or any combination thereof. In embodiments, the polymer is D-1184 (Kraton Polymers), D-1189 (Kraton Polymers), D-1191 (Kraton Polymers), G-1650 (Kraton Polymers), Elvax 240W (Dow), or any combination thereof. According to embodiments of this disclosure, a polymer is used in combination with a crude tall oil-based continuous phase material and a resinous hardening agent to further increase the softening point of the combination thereof and to reduce brittleness such that the three-way combination is suitable for use in a roofing material.

In an embodiment, a combination of polymers is used. In an embodiment, a combination of SBS and SEBS is used. In some embodiments, the ratio of SBS to SEBS is 1:1 to 10:1. In other embodiments, the ratio of SBS to SEBS is 1:1 to 8:1. In other embodiments, the ratio of SBS to SEBS is 1:1 to 6:1. In other embodiments, the ratio of SBS to SEBS is 1:1 to 4:1. In other embodiments, the ratio of SBS to SEBS is 1:1 to 2:1. In other embodiments, the ratio of SBS to SEBS is 2:1 to 10:1. In other embodiments, the ratio of SBS to SEBS is 2:1 to 8:1. In other embodiments, the ratio of SBS to SEBS is 2:1 to 6:1. In other embodiments, the ratio of SBS to SEBS is 4:1 to 10:1. In other embodiments, the ratio of SBS to SEBS is 4:1 to 8:1. In other embodiments, the ratio of SBS to SEBS is 4:1 to 6:1. In other embodiments, the ratio of SBS to SEBS is 6:1 to 10:1. In other embodiments, the ratio of SBS to SEBS is 6:1 to 8:1. In other embodiments, the ratio of SBS to SEBS is 8:1 to 10:1. In other embodiments, the ratio of SBS to SEBS is 5:1.

In an embodiment, the filled coating formulation comprises 1-10 wt % of polymer. In an embodiment, the filled coating formulation comprises 1-8 wt % of polymer. In an embodiment, the filled coating formulation comprises 1-6 wt % of polymer. In an embodiment, the filled coating formulation comprises 1-4 wt % of polymer. In an embodiment, the filled coating formulation comprises 1-2 wt % of polymer. In an embodiment, the filled coating formulation comprises 2-10 wt % of polymer. In an embodiment, the filled coating formulation comprises 2-8 wt % of polymer. In an embodiment, the filled coating formulation comprises 2-6 wt % of polymer. In an embodiment, the filled coating formulation comprises 2-4 wt % of polymer. In an embodiment, the filled coating formulation comprises 4-10 wt % of polymer. In an embodiment, the filled coating formulation comprises 4-8 wt % of polymer. In an embodiment, the filled coating formulation comprises 4-6 wt % of polymer. In an embodiment, the filled coating formulation comprises 6-10 wt % of polymer. In an embodiment, the filled coating formulation comprises 6-8 wt % of polymer. In an embodiment, the filled coating formulation comprises 8-10 wt % of polymer.

In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.01 to 1.25. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material

and resinous hardening agent ranges from 0.01 to 1. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.01 to 0.75. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.01 to 0.5. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.01 to 0.25. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.01 to 0.12. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.01 to 0.06. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.01 to 0.03. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.03 to 1.25. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.03 to 1. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.03 to 0.75. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.03 to 0.5. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.03 to 0.25. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.03 to 0.06. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.03 to 0.06. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.06 to 1.25. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.06 to 1. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.06 to 0.75. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.06 to 0.5. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.06 to 0.25. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.06 to 0.12. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.12 to 1.25. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.12 to 1. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.12 to 0.75. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.12 to 0.5. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.12 to 0.25. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.25 to 1.25. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent ranges from 0.25 to 1. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent

[illegible]

embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent is less than 0.11. In an embodiment, a ratio of polymer to crude tall oil-based continuous phase material and resinous hardening agent is less than 0.1.

Non-limiting examples of fillers used in the present disclosure include, without limitation, an organic filler, an inorganic mineral filler, and any combination thereof. In an embodiment, the filler is calcium carbonate, barium sulfate, calcium sulfate, talc, limestone, perlite, silica, fumed silica, precipitated silica, quartz, aluminum trihydrate, magnesium hydroxide, colemanite (e.g., hydrated calcium borate), titanium dioxide, fly ash, graphene nanoparticles, carbon black, recycled rubber tires, recycled shingles, recycled thermoplastic resins, basalt, roofing granules, clay, lignin, or any combination thereof. In an embodiment, the filler includes a high aspect ratio filler such as, e.g., graphene nanoparticles or carbon black. In an embodiment, the filler is a recycled material, such as post-consumer recycled asphalt shingles (PCRAS), ground tire rubber (GTR), acrylonitrile rubber (NBR), acrylonitrile butadiene styrene rubber (ABS), or other recycled thermoplastic(s). A non-limiting example of GTR includes GTR available from Lehigh Technologies, Tucker, GA.

In an embodiment, the filled coating formulation comprises a remainder of filler; in other words, the filled coating formulation comprises a crude tall oil-based continuous phase material, a resinous hardening agent, a polymer, and a filler in amounts such that the total wt % of the filled coating formulation is 100 wt %. In an embodiment, the filled coating formulation comprises 15-91 wt % of filler. In an embodiment, the filled coating formulation comprises 40-95 wt % of filler. In an embodiment, the filled coating formulation comprises 40-85 wt % of filler. In an embodiment, the filled coating formulation comprises 40-75 wt % of filler. In an embodiment, the filled coating formulation comprises 40-65 wt % of filler. In an embodiment, the filled coating formulation comprises 40-55 wt % of filler. In an embodiment, the filled coating formulation comprises 40-45 wt % of filler. In an embodiment, the filled coating formulation comprises 50-95 wt % of filler. In an embodiment, the filled coating formulation comprises 50-85 wt % of filler. In an embodiment, the filled coating formulation comprises 50-75 wt % of filler. In an embodiment, the filled coating formulation comprises 50-65 wt % of filler. In an embodiment, the filled coating formulation comprises 50-55 wt % of filler. In an embodiment, the filled coating formulation comprises 60-95 wt % of filler. In an embodiment, the filled coating formulation comprises 60-85 wt % of filler. In an embodiment, the filled coating formulation comprises 60-75 wt % of filler. In an embodiment, the filled coating formulation comprises 60-65 wt % of filler. In an embodiment, the filled coating formulation comprises 70-85 wt % of filler. In an embodiment, the filled coating formulation comprises 70-75 wt % of filler. In an embodiment, the filled coating formulation comprises 50 wt %, 60 wt %, 70 wt % or 80 wt % of filler.

Other components may also be added to the filled coating formulation to further modify or to provide additional properties, e.g., electrical conductivity. In an embodiment, the filled coating formulation further comprises a dye, a pigment, a fire retardant, a UV stabilizer, a tackifier, titanium dioxide, or any combination thereof. One of ordinary skill in the art would readily appreciate that such components could be included in a filled coating formulation in any amount suitable to achieve the purpose of its inclusion. Non-limiting examples of pigments and/or dyes include colorants, IR

reflective pigments and/or dyes, and phosphorescence and/or fluorescence pigments and/or dyes. Non-limiting examples of pigments include, but are not limited to, color pigments and/or reflective pigments, such as Colonial Red, which is a reflective pigment that is available from Americhem Inc., Cuyahoga Falls, Ohio. Non-limiting examples of UV stabilizers include, but are not limited to, UV absorbers, hindered amine light stabilizers, anti-oxidant pigments and/or carriers, such as PP, PE, or IPP.

In an embodiment, the coating does not comprise asphalt (i.e., is "free of asphalt"). In other words, the coating includes 0% by weight of asphalt. In an embodiment, the coating comprises 0.1% by weight to 49% by weight of asphalt. In an embodiment, the coating comprises 1% by weight to 35% by weight of asphalt. In an embodiment, the coating comprises 10% by weight to 25% by weight of asphalt.

In an embodiment, the filled coating formulation has a softening point ranging from 175-320° F. In an embodiment, the filled coating formulation has a softening point ranging from 175-300° F. In an embodiment, the filled coating formulation has a softening point ranging from 175-275° F. In an embodiment, the filled coating formulation has a softening point ranging from 175-250° F. In an embodiment, the filled coating formulation has a softening point ranging from 175-225° F. In an embodiment, the filled coating formulation has a softening point ranging from 175-200° F. In an embodiment, the filled coating formulation has a softening point ranging from 200-320° F. In an embodiment, the filled coating formulation has a softening point ranging from 200-300° F. In an embodiment, the filled coating formulation has a softening point ranging from 200-275° F. In an embodiment, the filled coating formulation has a softening point ranging from 200-250° F. In an embodiment, the filled coating formulation has a softening point ranging from 200-225° F. In an embodiment, the filled coating formulation has a softening point ranging from 225-320° F. In an embodiment, the filled coating formulation has a softening point ranging from 225-300° F. In an embodiment, the filled coating formulation has a softening point ranging from 225-275° F. In an embodiment, the filled coating formulation has a softening point ranging from 225-250° F. In an embodiment, the filled coating formulation has a softening point ranging from 250-320° F. In an embodiment, the filled coating formulation has a softening point ranging from 250-300° F. In an embodiment, the filled coating formulation has a softening point ranging from 250-275° F. In an embodiment, the filled coating formulation has a softening point ranging from 275-320° F. In an embodiment, the filled coating formulation has a softening point ranging from 275-300° F. In an embodiment, the filled coating formulation has a softening point ranging from 300-320° F.

In an embodiment, the filled coating formulation has a penetration ranging from 5-100 dmm at 77° F. In an embodiment, the filled coating formulation has a penetration ranging from 5-80 dmm at 77° F. In an embodiment, the filled coating formulation has a penetration ranging from 5-60 dmm at 77° F. In an embodiment, the filled coating formulation has a penetration ranging from 5-40 dmm at 77° F. In an embodiment, the filled coating formulation has a penetration ranging from 5-20 dmm at 77° F. In an embodiment, the filled coating formulation has a penetration ranging from 25-100 dmm at 77° F. In an embodiment, the filled coating formulation has a penetration ranging from 25-80 dmm at 77° F. In an embodiment, the filled coating formulation has a penetration ranging from 25-60 dmm at 77° F. In an embodiment, the filled coating formulation has a penetration

In an embodiment, the non-filled coating formulation comprises 20-90 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 20-80 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 20-70 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 20-60 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 20-50 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 20-40 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 20-30 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 30-90 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 30-80 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 30-70 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 30-60 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 30-50 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 30-40 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 40-90 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 40-80 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 40-70 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 40-60 wt % of crude tall oil-based continuous phase material. In an embodiment, the non-filled coating formulation comprises 40-50 wt % of crude tall

In an embodiment, the non-filled coating formulation comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 200-12,500 cP measured at 400° F. In an embodiment, the non-filled coating formulation comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 200-10,000 cP

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In an embodiment, the non-filled coating formulation has a softening point ranging from 145-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-250° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-225° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-200° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-175° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-150° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-250° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-225° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-200° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-175° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 175-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 175-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 175-250° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 175-225° F. In an embodiment,

the non-filled coating formulation has a softening point ranging from 175-200° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 200-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 200-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 200-250° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 225-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 225-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 225-250° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 250-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 250-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 275-305° F.

In an embodiment, the non-filled coating formulation has a penetration ranging from 15-150 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-110 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-90 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-70 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-50 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-30 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-150 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-110 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-90 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-70 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-50 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 55-150 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 55-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 55-110 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 55-90 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 55-70 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 75-150 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 75-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 75-110 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 75-90 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 95-150 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 95-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 95-110 dmm at 77° F. In an embodi-

ment, the non-filled coating formulation has a penetration ranging from 115-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 135-150 dmm at 77° F.

Another embodiment of the disclosure provides a non-asphaltic roofing shingle comprising a substrate having a coating which comprises a crude tall oil-based continuous phase material, a resinous hardening agent, a polymer, and a filler, wherein the softening point of the coating ranges from 175-320° F., and wherein the penetration of the coating ranges from 5-100 dmm measured at 77° F.

The coating in this embodiment is based on the filled coating formulation described above. Details regarding crude tall oil-based continuous phase materials, resinous hardening agents, polymers and additional components, as well as inclusion amounts and softening point and penetration ranges, are as set forth above for the filled coating formulation.

In an embodiment, the substrate comprises a fiberglass mat, a polyester mat, a scrim, a coated scrim, or a combination thereof. In an embodiment, the substrate comprises a synthetic or natural scrim. In some embodiments, the substrate or mat includes nano-fibrillated cellulose fibers.

In an embodiment, the non-asphaltic roofing shingle is configured to be prepared on a substantially standard manufacturing line for asphaltic shingles at a standard speed, ranging from 110 feet per minute (FPM) to 1000 feet per minute (FPM). A non-limiting example of a substantially standard manufacturing line for asphaltic shingles is detailed in U.S. Pat. No. 10,195,640, the contents of which are hereby incorporated by reference.

In an embodiment, the non-asphaltic roofing shingle is a single layer shingle or a laminated shingle having two or more layers.

In an embodiment, the non-asphaltic roofing shingle satisfies ICC acceptance criteria for an alternative non-asphaltic shingle.

In an embodiment, the non-asphaltic roofing shingle further comprises granules. In an embodiment, granules are applied to a surface of the non-asphaltic roofing shingle. In an embodiment, the non-asphaltic roofing shingle includes mineral surfacing, such as, e.g., fines, granules, sand, metal flakes and/or reflective granules. In an embodiment, the non-asphaltic roofing shingle includes polymer films and/or synthetic and/or natural non-woven and/or woven fabrics, with or without decorative elements, including, for example, printing, embossing and/or protective coatings, on the coating. In an embodiment, photo (e.g., UV) and/or thermal stabilizers are added to a surface of the coating and/or non-asphaltic roofing shingle.

In an embodiment, a thickness of the coating on the substrate ranges from 5-100 mils, from 10-80 mils, or from 15-40 mils.

In an embodiment, the non-asphaltic roofing shingle comprises one or more layers of the coating, discussed above. In some embodiments, there are one to fifteen layers of the coating. In some embodiments, there are one to twelve layers of the coating. In some embodiments, there are one to ten layers of the coating. In some embodiments, there are one to eight layers of the coating. In some embodiments, there are one to five layers of the coating. In some embodiments, there are one to three layers of the coating. In some embodiments, there are three to fifteen layers of the coating. In some embodiments, there are five to fifteen layers of the coating. In some embodiments, there are eight to fifteen layers of the coating. In some embodiments, there are ten to

fifteen layers of the coating. In some embodiments, there are twelve to fifteen layers of the coating.

In an embodiment, the coating comprises at least one layer that is applied to both a top surface and a bottom surface of the substrate.

Another embodiment of the disclosure provides a non-asphaltic roofing shingle comprising a substrate having a non-filled coating which comprises a crude tall oil-based continuous phase material, a resinous hardening agent, and a polymer, wherein the softening point of the coating ranges from 145-305° F., and wherein the penetration of the coating ranges from 15-150 dmm measured at 77° F.

The coating in this embodiment is based on the non-filled coating formulation described above. Details regarding crude tall oil-based continuous phase materials, resinous hardening agents, polymers and additional components, as well as inclusion amounts and softening point and penetration ranges, are as set forth above for the non-filled coating formulation. In addition, details regarding the substrate, manufacturing, and structure of the non-asphaltic roofing shingle are as set forth above for the non-asphaltic roofing shingle coated with the filled coating formulation.

Additional embodiments of the disclosure provide other roofing materials such as underlayment, modified bitumen roofing ("mod bit"), rolled product, roofing membrane, etc., any of which comprising the filled coating formulation or the non-filled coating formulation described above.

Another embodiment of the disclosure provides a method comprising: obtaining a substrate; and coating the substrate with a filled coating formulation to form a non-asphaltic roofing material, wherein the filled coating formulation comprises 5-40 wt % of a crude tall oil-based continuous phase material; 3-35 wt % of a resinous hardening agent; 1-10 wt % of a polymer; and a filler, and wherein a viscosity of the filled coating formulation ranges from 2000-20000 cP measured at 400° F.

The coating in this embodiment is based on the filled coating formulation described above. Details regarding crude tall oil-based continuous phase materials, resinous hardening agents, polymers and additional components, as well as inclusion amounts, softening point, and penetration, are as set forth above for the filled coating formulation. In addition, details regarding the substrate, as well as the characteristics of the non-asphaltic roofing material resulting from coating the substrate with the filled coating formulation may be as set forth above for the non-asphaltic roofing shingle coated with the filled coating formulation.

In an embodiment, coating the substrate is performed on a substantially standard manufacturing line for asphaltic shingles at a standard speed, ranging from 110 feet per minute (FPM) to 1000 feet per minute (FPM). As noted above, a non-limiting example of a substantially standard manufacturing line for asphaltic shingles is detailed in U.S. Pat. No. 10,195,640.

In some embodiments of the method, one or more additional manufacturing steps are carried out. Some embodiments of the method further comprise applying granules, applying a polymer film or fabric, and/or photo and/or thermal stabilizers to a surface of the non-asphaltic roofing shingle. Details regarding the granules, polymer film or fabric, and photo and/or thermal stabilizers are as set forth above for the non-asphaltic roofing shingle coated with the filled coating formulation. In an embodiment, the method further comprises applying one or more layers of the coating to form a non-asphaltic roofing material.

In an embodiment, the method further comprises preparing the coating. In an embodiment, the coating is prepared

by mixing the components using static mixing, a low shear mixer, and/or a high shear mixer. A non-limiting example of a low shear mixer is EUROSTAR® 60 Digital, IKA Works, Inc., Wilmington, NC, which mixes batches at about 500 to 1500 RPM, with a paddle-type blade to generate low shear. A non-limiting example of a high shear mixer is SILVERSON® L5M-A Laboratory Mixer, Silverson Machines, Inc., East Longmeadow, MA, which mixes batches at or above 1000 RPM, with a blade and a head that are configured to generate high shear, as well as heat mixing.

In an embodiment, the coating is in the form of a pourable coating formulation that is poured onto a substrate on one or both sides and roll pressed to impregnate and saturate the substrate. In an embodiment, granules are then applied. Some embodiments of the method further comprise cutting the formed non-asphaltic roofing material to provide one or more roofing shingle layers and configuring the one or more layers to provide a shingle having the desired structure.

Another embodiment of the disclosure provides a method comprising: obtaining a substrate; and coating the substrate with a non-filled coating formulation to form a non-asphaltic roofing material, wherein the non-filled coating formulation comprises 20-90 wt % of a crude tall oil-based material, 10-70 wt % of a resinous hardening agent, and 5-20 wt % of a polymer, and wherein a viscosity of the coating formulation ranges from 200-12,500 cP measured at 400° F.

The coating in this embodiment is based on the non-filled coating formulation described above. Details regarding crude tall oil-based continuous phase materials, resinous hardening agents, polymers and additional components, as well as the inclusion amounts, softening point and penetration, are as set forth above for the non-filled coating formulation. In addition, details regarding the substrate, as well as the characteristics of the non-asphaltic roofing material resulting from coating the substrate with the non-filled coating formulation may be as set forth above for the non-asphaltic roofing shingle coated with the non-filled coating formulation. Details regarding manufacturing steps are as set forth above for the method which employs the filled coating formulation, though adjustments may be made to account for viscosity differences.

The disclosure provides additional alternative embodiments in which the crude tall oil-based continuous phase material is replaced by or used in conjunction with a plant- or bio-derived continuous phase material. One such alternative embodiment of the disclosure provides a non-filled coating formulation comprising:

20-60 wt % of a plant- or bio-derived continuous phase material;

30-75 wt % of a resinous hardening agent; and
5-20 wt % of a polymer,

wherein a viscosity of the non-filled coating formulation ranges from 200-12500 cP measured at 400° F.

Non-limiting examples of plant- or bio-derived continuous phase materials used in the present disclosure include, without limitation, corn oil, vegetable oil, triglycerides, renewable oil technology, and combinations thereof. In an embodiment, the plant- or bio-derived continuous phase material is Anova 1815 (Cargill), low viscosity vegetable oil (Cargill), renewable oil technology based on triglycerides (such as Tufftek 4002) or any combination thereof. As used herein, "plant- or bio-based continuous phase material(s)" refers to any continuous phase material derived from a plant- or bio-source, including chemically modified or functionalized versions thereof for purposes of the present disclosure, the plant- or bio-derived continuous phase material is not a crude tall oil-based continuous phase material as defined

In an embodiment, the non-filled coating formulation comprising a plant- or bio-derived continuous phase material comprises 30-75 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 30-70 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 30-65 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 30-60 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 30-55 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 30-50 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 30-45 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 30-40 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 30-35 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprising a plant- or bio-derived continuous phase material comprises 35-75 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 35-70 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 35-65 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 35-60 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 35-55 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 35-50 wt % of resinous hardening agent. In an embodiment, the non-filled coating formulation comprises 35-45 wt % of resinous hardening agent. In an

60 In an embodiment, the non-filled coating formulation comprising a plant- or bio-derived continuous phase material comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging from 200-12,500 cP measured at 400° F. In an embodiment, the non-filled
65 coating formulation comprising a plant- or bio-derived continuous phase material comprises an amount of resinous hardening agent sufficient to achieve a viscosity ranging

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In an embodiment, the non-filled coating formulation comprising a plant- or bio-derived continuous phase material comprises 5-20 wt % of polymer. In an embodiment, the non-filled coating formulation comprises 5-17 wt % of polymer. In an embodiment, the non-filled coating formulation comprises 5-14 wt % of polymer. In an embodiment, the non-filled coating formulation comprises 5-11 wt % of polymer. In an embodiment, the non-filled coating formulation comprises 5-8 wt % of polymer. In an embodiment, the non-filled coating formulation comprising a plant- or bio-derived continuous phase material comprises 8-20 wt % of polymer. In an embodiment, the non-filled coating formulation comprises 8-17 wt % of polymer. In an embodiment, the non-filled coating formulation comprises 8-14 wt % of polymer. In an embodiment, the non-filled coating formulation comprises 8-11 wt % of polymer. In an embodiment, the non-filled coating formulation comprising a plant- or bio-derived continuous phase material comprises 11-20 wt % of polymer. In an embodiment, the non-filled coating formulation comprises 11-17 wt % of polymer. In an embodiment, the non-filled coating formulation comprises 11-14 wt % of polymer. In an embodiment, the non-filled coating formulation comprising a plant- or bio-derived continuous phase material comprises 14-20 wt % of polymer. In an embodiment, the non-filled coating formulation comprising

prises 14-17 wt % of polymer. In an embodiment, the non-filled coating formulation comprises 17-20 wt % of polymer.

In an embodiment, the non-filled coating formulation comprising a plant- or bio-derived continuous phase material has a softening point ranging from 145-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-250° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-225° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-200° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-175° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 145-150° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-250° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-225° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-200° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 150-175° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 175-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 175-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 175-250° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 175-225° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 175-200° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 200-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 200-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 200-250° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 200-225° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 225-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 225-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 225-250° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 250-305° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 250-275° F. In an embodiment, the non-filled coating formulation has a softening point ranging from 275-305° F.

In an embodiment, the non-filled coating formulation comprising a plant- or bio-derived continuous phase material has a penetration ranging from 15-150 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-110 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-90 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-70 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-50 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 15-30 dmm at

77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-150 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-110 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-90 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-70 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 35-50 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 55-150 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 55-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 55-110 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 55-90 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 55-70 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 75-150 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 75-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 75-110 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 75-90 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 95-150 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 95-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 95-110 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 115-150 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 115-130 dmm at 77° F. In an embodiment, the non-filled coating formulation has a penetration ranging from 135-150 dmm at 77° F.

Further alternative embodiments of the disclosure include a filled coating formulation comprising a plant- or bio-derived continuous phase material, a resinous hardening agent, a polymer, and a filler; a non-asphaltic roofing shingle comprising a substrate having a coating which comprises a plant- or bio-derived continuous phase material, a resinous hardening agent, a polymer, and a filler; and a method comprising obtaining a substrate and coating the substrate with a filled coating formulation to form a non-asphaltic roofing material, wherein the filled coating formulation comprises a plant- or bio-derived continuous phase material, a resinous hardening agent, a polymer, and a filler.

EXAMPLES

Specific embodiments of the invention will now be demonstrated by reference to the following examples. It should be understood that these examples are disclosed by way of illustrating the disclosure and should not be taken in any way to limit the scope of the present disclosure.

Examples 1 and 2 and Comparative Examples A-D

Blends of different crude tall oil-based continuous phase materials and resinous hardening agents were made and characterized as set forth in Table 1 below:

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TABLE 1

Softening Points of Tall Oil Pitch/Rosin Ester Blends.			
Tall oil pitch (TOP)	Rosin ester	Ratio of TOP/rosin ester (w/w)	Softening point (SP), ° F.
SylfatDP-1		1:0	<77
Sylfat DP-8		1:0	<77
SylfatDP-1	Sylvatac RE-98	1:1	113
SylfatDP-1	Sylvacote 7097	1:1	164
SylfatDP-1	Dertoline P110	1:1	119
Sylfat DP-8	Sylvacote 7097	1:1	125

The addition of a resinous hardening agent to a crude tall oil-based continuous phase material increased the softening point. However, in order to obtain a sufficiently flexible material, as well as meet other property requirements, a polymer component was also added as set forth in Table 2 to form non-filled coating formulations according to the present disclosure.

TABLE 2

Physical Properties of Formulations Acceptable as Roofing Materials.			
Formulation No.			
	Example 1	Example 2	
TOP, wt %	DP-1, 44 wt %	DP-8, 44 wt %	
Rosin ester, wt %	P110, 44 wt %	P110, 44 wt %	
SBS, wt %	D-1184, 10 wt %	D-1184, 10 wt %	
SEBS, wt %	G-1650, 2 wt %	G-1650, 2 wt %	
SP, ° F.	206	242	
Penetration, dmm at 77° F.	54	60	
Viscosity, cP at 400° F.	947	1307	

The omission of the rosin ester in a formulation leads to material property deficiencies. This is shown in Table 3 below.

TABLE 3

Comparison of Formulation Physical Properties with and without Rosin Esters.					
Formulation No.					
	Example 2	Comparative Example A	Comparative Example B	Comparative Example C	Comparative Example D
TOP, wt %	DP-8, 44 wt %	DP-8, 90 wt %	DP-8, 95 wt %	DP-8, 90 wt %	DP-8, 82 wt %
Rosin ester, wt %	P110, 44 wt %	—	—	—	—
SBS, wt %	D-1184, 10 wt %	D-1184, 8 wt %	D-1184, 5 wt %	D-1184, 10 wt %	D-1184, 18 wt %
SEBS, wt %	G-1650, 2 wt %	G-1650, 2 wt %	—	—	—
SP, ° F.	242	201	154	216	260
Penetration, dmm at 77° F.	60	200	Too soft to determine	144	102
Viscosity, cP at 400° F.	1307	188	Not tested	96	2611 (360° F.)

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In Comparative Example A, the softening point was acceptable, and the viscosity was comparable to asphalt; however, the material was very soft, as shown by the penetration data and had little mechanical integrity. Comparative Example B clearly did not have enough polymer to raise the softening point sufficiently and was too soft. Comparative Examples C and D had acceptable softening points; however, Comparative Example C was still very soft and had little mechanical strength, while Comparative Example D was very viscous.

Corresponding filled coating formulations were prepared according to the present disclosure by adding limestone filler to the non-filled coating formulations of Examples 1 and 2, as set forth in Table 4 below.

TABLE 4

Filled Non-Asphaltic Coating Formulations.			
Formulation No.			
	Example 1F, 70% filler content	Example 2F, 70% filler content	
Filler, wt %	Limestone (CaCO ₃), 70 wt %	Limestone (CaCO ₃), 70 wt %	
TOP, wt %	DP-1, 13.2 wt %	DP-8, 13.2 wt %	
Rosin ester, wt %	P110, 13.2 wt %	P110, 13.2 wt %	
SBS, wt %	D-1184, 3 wt %	D-1184, 3 wt %	
SEBS, wt %	G-1650, 0.6 wt %	G-1650, 0.6 wt %	

The resulting filled coating formulation properties are shown in Table 5 below, along with relevant properties of asphalt-based formulations.

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TABLE 5

Comparison of Filled Coating Properties.				
Formulation No.				
	Comparative Example E (blown asphalt coating with 65% filler content)	Comparative Example F (polymer modified asphalt coating with 68% filler content)	Example 1F	Example 2F
SP, ° F.	242	249	218	258
Penetration, dmm at 77° F.	9	15	18	27
Viscosity, cP at 400° F.	2418	3648	9472	10610

It can be seen that the non-asphaltic coatings according to the present disclosure, i.e., Examples 1F and 2F, have comparable softening points relative to asphaltic coatings and are generally softer (higher penetration values) and more viscous, but are manufacturable using conventional shingle manufacturing equipment.

Coupons of the filled formulations of Examples 1F and 2F were prepared for testing as Examples 1C and 2C. The resulting coupons prepared from Example 2F are shown in FIGS. 1 and 2. FIG. 1 shows the front (exposed) sides of the coupons, with a granulated surface for protection against the

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the right image showing the glass mat without coating. With regard to performance as a roofing shingle, test results are shown in Table 6 below.

TABLE 6

Shingle Coupon Test Results.				
Formulation No.				
	Comparative Example E	Comparative Example F	Example 1C	Example 2C
Tensile, MD (lb-f)	285	229	64	63
Tensile, CD (lb-f)	152	113	36	36
Tear, MD (g-f)	835	1363	2165	1217
Tear, CD (g-f)	1176	1907	2374	1417
Fastener Pull (lb-f)	25	Not tested	23	21
Granule loss (g)	Not tested	Not tested	0.22	0.71

These results show that coupons prepared from the non-asphaltic coatings of the present disclosure have lower tensile strengths compared to their asphalt-based analogs, but have comparable or improved tear strengths.

Examples 3-17

Additional non-filled coating formulations were prepared and characterized as set forth in Table 7 below.

TABLE 7

Variations of the Non-Filled Coating Formulations.								
Formulation No.	TOP (wt %)	Rosin Ester (wt %)	Polymer 1 (wt %)	Polymer 2 (wt %)	SP, ° F.	Viscosity, cP at 400° F.	Penetration, dmm at 77° F.	
Example 3	Poix de tall oil (44%)	Dertoline P110 (44%)	D-1184 (10%)	G-1650 (2%)	228	1612	19	
Example 4	Altapyne pitch (44%)	Dertoline P110 (44%)	D-1184 (10%)	G-1650 (2%)	215	1182	36	
Example 5	Sylfat DP-1 (66%)	Sylvacote 7097 (22%)	D-1184 (10%)	G-1650 (2%)	210	295	116	
Example 6	Sylfat DP-8 (44%)	Sylvatac RE-98 (44%)	D-1184 (10%)	G-1650 (2%)	210	2205	81	
Example 7	Sylfat DP-8 (44%)	Dertoline P110 (44%)	D-1184 (10%)	Elvax 240W (2%)	235	976	60	
Example 8	Sylfat DP-8 (45%)	Dertoline P110 (45%)	D-1184 (10%)	—	237	803	69	
Example 9	Sylfat DP-8 (44%)	Dertoline P110 (44%)	D-1189 (10%)	G-1650 (2%)	232	790	48	
Example 10	Sylfat DP-8 (44%)	Dertoline P110 (44%)	D-1184 (12%)	—	254	1603	50	
Example 11	Tufftrek 3100 (44%)	Sylvacote 7097 (44%)	D-1184 (12%)	—	254	624	58	
Example 12	Sylfat DP-8 (45%)	WestRez 5110 (45%)	D-1184 (10%)	—	227	683	51	
Example 13	Sylfat DP-8 (45%)	Sylvacote 4984 (45%)	D-1184 (10%)	—	241	421	39	
Example 14	Sylfat DP-8 (45%)	Dertoline P105 (45%)	D-1184 (10%)	—	222	750	56	
Example 15	Sylfat DP-1 (35%)	Sylvatac RE-98 (53%)	D-1184 (10%)	G-1650 (2%)	180	1014	61	
Example 16	Sylfat DP-1 (62%)	Sylvacote 7097 (26%)	D-1184 (10%)	G-1650 (2%)	237	345	83	
Example 17	Sylfat DP-1 (66%)	Sylvacote 7097 (22%)	D-1184 (10%)	G-1650 (2%)	210.5	295	116	

elements. FIG. 2 shows the back of the coupons, with the left image showing a back side coated with filled coating and surfaced with fines, similar to an asphalt-based shingle, and

Corresponding filled coating formulations were prepared according to the present disclosure by adding limestone (calcium carbonate) filler to the non-filled coating formula-

tions of Examples 3-5, 7-9, and 12-14. The resulting filled coating formulation properties are shown in Table 8 below, along with the respective filler contents.

TABLE 8

Physical Properties of the Filled Coatings.				
Formulation No.	Filler content (wt %)	SP, ° F.	Viscosity, cP at 400° F.	Penetration, dmm at 77° F.
Example 3F	65	236	9892	13
Example 4F	65	224	7040	18
Example 5F	70	245	5067	48
Example 7F	70	266	10430	22
Example 8F	65	251	5102	34
Example 9F	65	239	3861	28
Example 12F	65	226	3685	25
Example 13F	65	258	11800	17
Example 14F	65	232	4160	26

Coupons of the filled formulations of Examples 3F-5F, 7F-9F, and 12F-14F were prepared for testing as Examples 3C-5C, 7C-9C, and 12C-14C. Coupons consist of fiberglass mat impregnated with filled coating. Roofing granules are applied to the coupons for the rub loss tests. Otherwise, the coupons are tested without granules applied. With regard to performance as a roofing shingle, test results are shown in Table 9 below.

TABLE 9

Shingle Coupon Test Results.						
Formulation No.	Tensile, MD (lb-f)	Tensile, CD (lb-f)	Tear, MD (g-f)	Tear, CD (g-f)	Nail Pull (lb-f)	Rub loss (g)
Example 3C	80.1	not tested	not tested	1594	not tested	not tested
Example 4C	69.9	not tested	not tested	1720	not tested	not tested
Example 5C	39.2	not tested	not tested	1511	12.6	2.43
Example 7C	56.0	not tested	not tested	1558	not tested	not tested
Example 8C	60.0	not tested	not tested	1428	not tested	not tested
Example 9C	57.5	not tested	not tested	1328	not tested	not tested
Example 12C	85.9	53.7	952	1301	24.9	0.90
Example 13C	69.5	35.0	1030	1362	35.1	1.79
Example 14C	82.4	45.4	1123	1275	25.2	0.84

Examples 18-21

Additional non-filled coating formulations were prepared and characterized as set forth in Table 10 below.

TABLE 10

Variations of the Non-Filled Coating Formulations.							
Formulation No.	Plant-derived continuous phase material (wt %)	Rosin Ester (wt %)	Polymer 1 (wt %)	Polymer 2 (wt %)	SP, ° F.	Viscosity, cP at 400° F.	Penetration, dmm at 77° F.
Example 18	Cargill Anova 1815 (44.4%)	Sylvacote 7097 (41.6%)	D-1184 (12%)	G-1650 (2%)	182.2	269.7	86
Example 19	Cargill Anova 1815 (44%)	Sylvacote 7097 (42%)	D-1191 (14%)		224.3	369.1	93.5
Example 20	Cargill low viscosity vegetable oil (29%)	Sylvares 115 (61%)	D-1191 (7%)	Polystyrene from styrofoam (3%)	162.7	337.5	23.3
Example 21	Tufftrek 4002 (35%)	Sylvacote 7097 (53%)	D-1184 (10%)	G-1650 (2%)	242	293	76

List and Description of Materials Used

Name	Manufacturer	Description	CAS No.
Dertoline Poix de Tall Oil	Les Derives Resiniques et Terpeniques (DRT)	Tall oil pitch	8016-81-7
Altapyne Pitch	Ingevity Corporation	Tall oil pitch	8016-81-7
Sylfat DP1	Kraton Chemical LLC	Tall oil pitch blend	8016-81-7
Sylfat DP8	Kraton Chemical LLC	Tall oil pitch blend; low sterols	none
Anova 1815	Cargill	Corn oil	
187-1740 low viscosity modifier	Cargill	Low viscosity vegetable oil	
D-1184	Kraton Polymers	Branched SBS block copolymer, 30% styrene	9003-55-8
D-1189	Kraton Polymers	Radial SBS block copolymer, 32% styrene	9003-55-8
G-1650	Kraton Polymers	Linear SEBS block copolymer, 29% styrene	66070-58-4
Elvax 240W	Dow	ethylene-vinyl acetate copolymer,	24937-78-8

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-continued

List and Description of Materials Used			
Name	Manufacturer	Description	CAS No.
Dertoline P110	Les Derives Resiniques et Terpeniques (DRT)	28% vinyl acetate Esters of maleated rosin with pentaerythritol	94581-17-6
Sylvatac RE-98	Kraton Chemical LLC	Resin acids and rosin acids, esters with pentaerythritol	8050-26-8
Sylvacote 7097	Kraton Chemical LLC	Rosin, maleated, polymer with glycerol	68038-41-5
Tufftrek 3100	Georgia Pacific Chemicals	Crude tall oil	
Tufftrek 4002	Georgia Pacific	Renewable oil technology (based on triglycerides)	
WestRez 5110 Dertoline P105	Ingevity Corporation Les Derives Resiniques et Terpeniques (DRT)	Modified rosin ester pentaerythritol ester of rosin	proprietary 8050-26-8
Sylvacote 4984	Kraton Chemical LLC	Rosin, maleated, polymer with glycerol	68038-41-5
D-1191	Kraton Polymers	Radial SBS Polystyrene from Styrofoam	9003-55-8

Although the disclosure has been described in certain specific exemplary embodiments, many additional modifications and variations would be apparent to those skilled in the art in light of this disclosure. It is, therefore, to be understood that this disclosure may be practiced otherwise than as specifically described. Thus, the exemplary embodiments of the disclosure should be considered in all respects to be illustrative and not restrictive, and the scope of the disclosure to be determined by any claims supportable by this disclosure and the equivalents thereof, rather than by the foregoing description.

We claim:

1. A roofing material comprising a substrate having a coating that comprises:

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(a) 40-50 wt % of a plant-or bio-derived continuous phase material;

(b) 40-50 wt % of a resinous hardening agent; and

(c) 5-20 wt % of a polymer;

wherein a ratio of the plant-or bio-derived continuous phase material to the resinous hardening agent is about 1:1, based on wt %;

wherein a viscosity of the coating ranges from 200-12,500 cP measured at 400° F.; and

wherein the roofing material is selected from the group consisting of an underlayment, modified bitumen roofing, a rolled product, and a roofing membrane.

2. The roofing material according to claim 1, wherein the plant-or bio-derived continuous phase material comprises corn oil, vegetable oil, triglycerides, or a combination thereof.

3. The roofing material according to claim 1, wherein the resinous hardening agent comprises esters of maleated rosin with pentaerythritol; rosin, maleated, polymer with glycerol; alpha methyl styrene tall oil resin; or a combination thereof.

4. The roofing material according to claim 1, wherein the polymer comprises poly (styrene-butadiene-styrene) (SBS), poly (styrene-ethylenebutylene-styrene) (SEBS), ethylene-vinyl acetate copolymer (EVA), polyvinyl butyral (PVB), or a combination thereof.

5. The roofing material according to claim 1, wherein the coating further comprises a dye, a pigment, a fire retardant, a UV stabilizer, or a combination thereof.

6. The roofing material according to claim 1, wherein the substrate comprises a fiberglass mat, a polyester mat, a scrim, a coated scrim, or a combination thereof.

7. The roofing material according to claim 1, wherein a thickness of the coating on the substrate ranges from 5-100 mils.

8. The roofing material according to claim 1, wherein the roofing material comprises one or more layers of the coating.

9. The roofing material according to claim 1, wherein the coating does not comprise asphalt.

10. The roofing material according to claim 1, wherein the polymer comprises poly (styrene-butadiene-styrene) (SBS), polyvinyl butyral (PVB), or a combination thereof.

11. The roofing material according to claim 1, wherein the coating comprises 40-45 wt % of the resinous hardening agent.

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